

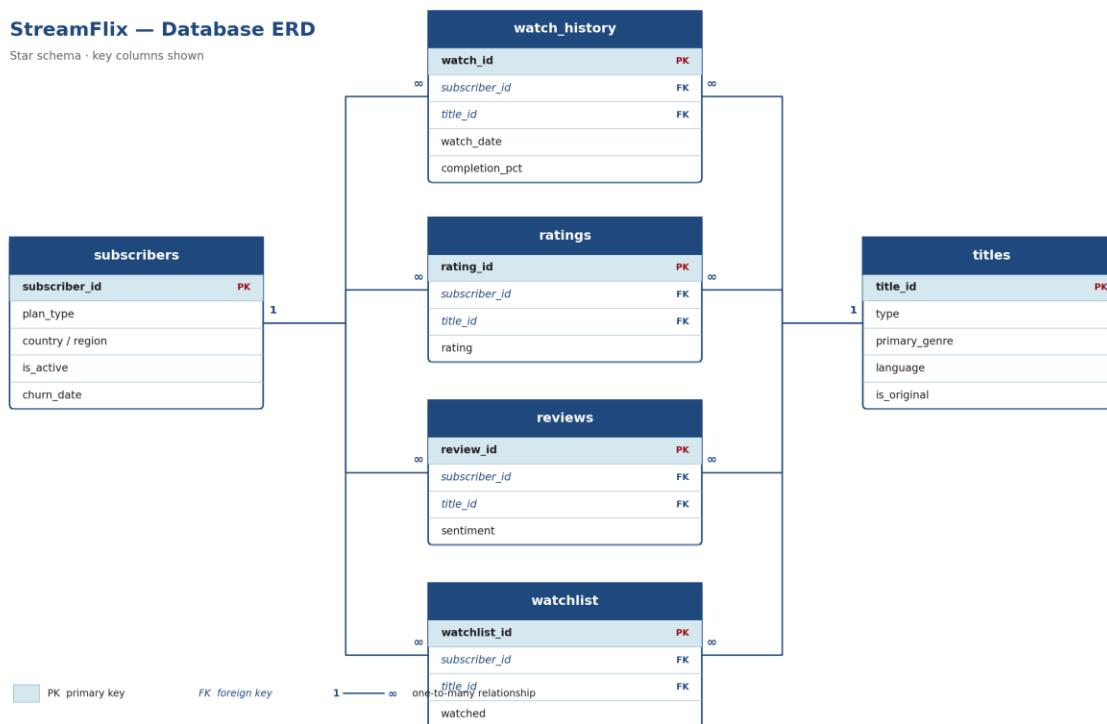
StreamFlix — Data Dictionary

Companion reference for the Content Analytics Project

This document defines every table and column in the StreamFlix dataset. The data is organised as a star schema: watch_history is the central fact table, with subscribers and titles as the two main dimension tables. The ratings, reviews and watchlist tables each link a subscriber to a title.

1. Entity-Relationship Diagram

The diagram below shows the six tables and how they join. Every event table links back to both subscribers (via subscriber_id) and titles (via title_id).



2. Tables at a Glance

Table	Grain (1 row =)	Rows	Primary Key
subscribers	one subscriber	15,000	subscriber_id
titles	one movie or series	9,000	title_id
watch_history	one viewing session	650,000	watch_id
ratings	one star rating	130,000	rating_id
reviews	one written review	110,000	review_id
watchlist	one saved title	65,000	watchlist_id

3. Column Definitions

3.1 subscribers

Column	Type	Key	Description
subscriber_id	VARCHAR	PK	Unique subscriber identifier
signup_date	DATE		Date the subscriber joined StreamFlix
country	VARCHAR		Subscriber's country (20 values)
region	VARCHAR		World region derived from country (9 values)
age	INT		Subscriber age in years (13–85)
gender	VARCHAR		Female / Male / Non-binary / Undisclosed
plan_type	VARCHAR		Basic with Ads / Standard / Premium
monthly_price_usd	DECIMAL		Monthly subscription price (6.99 / 15.49 / 22.99)
household_size	INT		Number of people in the household
primary_device	VARCHAR		Device most used by the subscriber
payment_method	VARCHAR		Credit Card / Debit Card / PayPal / Mobile Wallet / Gift Card / Carrier Billing
tenure_months	INT		Months since signup (as of 2026-06-01)
is_active	BOOLEAN		Whether the account is currently active
churn_date	DATE		Date of churn; blank if still active

3.2 titles

Column	Type	Key	Description
title_id	VARCHAR	PK	Unique title identifier
title_name	VARCHAR		Name of the movie or series
type	VARCHAR		Movie or TV Show
primary_genre	VARCHAR		Main genre (16 values)
country	VARCHAR		Country of origin (20 values)
language	VARCHAR		Primary language (14 values)
release_year	INT		Year the title was released (1969–2026)
date_added	DATE		Date the title was added to the catalogue
maturity_rating	VARCHAR		Content rating (e.g. PG, TV-MA, R)
seasons	INT		Number of seasons; 0 for movies
content_duration_min	INT		Runtime per play in minutes
is_original	BOOLEAN		Whether it is a StreamFlix Original
license_type	VARCHAR		Original / Exclusive License / Non-Exclusive License / Co-Production
director	VARCHAR		Director name

Column	Type	Key	Description
cast	VARCHAR		Comma-separated principal cast
quality_score	DECIMAL		Internal quality score (0–100)
popularity_score	DECIMAL		Relative popularity weight (0–1)
license_cost_usd	DECIMAL		Cost to license or produce the title
license_expiry	DATE		Licence expiry date; blank for Originals
total_watch_hours	DECIMAL		Total hours watched across all subscribers
total_plays	INT		Total viewing sessions started

3.3 watch_history (central fact table)

Column	Type	Key	Description
watch_id	BIGINT	PK	Unique viewing-session identifier
subscriber_id	VARCHAR	FK	Links to subscribers.subscriber_id
title_id	VARCHAR	FK	Links to titles.title_id
watch_date	DATE		Date of the viewing session
device	VARCHAR		Device used (6 values)
region	VARCHAR		Region where the session occurred (9 values)
content_duration_min	INT		Full length of the content in minutes
watch_duration_min	DECIMAL		Minutes actually watched in the session
completion_pct	DECIMAL		Percent of the content completed (0–100)
completed	BOOLEAN		Whether the session passed the completion threshold

3.4 ratings

Column	Type	Key	Description
rating_id	BIGINT	PK	Unique rating identifier
subscriber_id	VARCHAR	FK	Links to subscribers.subscriber_id
title_id	VARCHAR	FK	Links to titles.title_id
rating	INT		Star rating on a 1–5 scale
rating_date	DATE		Date the rating was given

3.5 reviews

Column	Type	Key	Description
review_id	BIGINT	PK	Unique review identifier
subscriber_id	VARCHAR	FK	Links to subscribers.subscriber_id
title_id	VARCHAR	FK	Links to titles.title_id
review_text	VARCHAR		Free-text review

Column	Type	Key	Description
sentiment	VARCHAR		Positive / Neutral / Negative
helpful_votes	INT		Number of helpful votes the review received
review_date	DATE		Date the review was posted

3.6 watchlist

Column	Type	Key	Description
watchlist_id	BIGINT	PK	Unique watchlist-entry identifier
subscriber_id	VARCHAR	FK	Links to subscribers.subscriber_id
title_id	VARCHAR	FK	Links to titles.title_id
added_date	DATE		Date the title was added to the watchlist
watched	BOOLEAN		Whether the saved title was later watched

4. Relationships

All relationships are one-to-many, flowing from the dimension tables to the event tables:

- subscribers (1) → (many) watch_history, ratings, reviews, watchlist — via subscriber_id
- titles (1) → (many) watch_history, ratings, reviews, watchlist — via title_id

Join example: `pd.merge(watch_history, titles, on='title_id', how='left')` links each session to its content metadata. Add a second merge on `subscriber_id` to bring in subscriber attributes.

5. Category Value Reference

Field	Allowed Values
plan_type	Basic with Ads, Standard, Premium
type	Movie, TV Show
primary_genre	Action, Animation, Comedy, Crime, Documentary, Drama, Fantasy, Horror, Kids & Family, Musical, Mystery, Reality, Romance, Sci-Fi, Stand-Up, Thriller
language	Arabic, English, French, German, Hindi, Indonesian, Italian, Japanese, Korean, Portuguese, Spanish, Swedish, Thai, Turkish
region	Africa, East Asia, Europe, Latin America, Middle East, North America, Oceania, South Asia, Southeast Asia
primary_device / device	Smart TV, Mobile, Laptop, Tablet, Game Console, Streaming Stick
payment_method	Credit Card, Debit Card, PayPal, Mobile Wallet, Gift Card, Carrier Billing
gender	Female, Male, Non-binary, Undisclosed
maturity_rating	G, PG, PG-13, R, NC-17, TV-Y, TV-G, TV-PG, TV-14, TV-MA
license_type	Original, Exclusive License, Non-Exclusive License, Co-Production
sentiment	Positive, Neutral, Negative

6. Notes for Analysts

- `total_watch_hours` and `total_plays` in titles are pre-aggregated from `watch_history` — use them for quick rollups, or recompute from the sessions for full control.
- Dates are stored in ISO format (YYYY-MM-DD). Parse with `pd.to_datetime()`.
- `churn_date` is populated only for inactive subscribers; treat it as an outcome, not a predictor.
- `seasons` is 0 for movies; `content_duration_min` is the length of a single play (episode or film).